Guneev Pannu

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Education

University of Toronto – 3.96 GPA Sept 2024 – May 2028

HBSc Double Major in Computer Science and Statistics, Math Minor

Relevant Courses: Data Structures & Algorithms, Operating Systems, Systems Programming, Databases, Security

Experience

Teaching Assistant Sept 2026 – Dec 2026

University of Toronto *Toronto, Canada*

- Leading weekly tutorials for **40+** students, Python fundamentals (control flow, functions, lists) for CSC108
- Authored weekly lab materials and slide decks in Latex to deepen understanding of **500+ students** for CSC108

Product Manager Sept 2026 – Present

UofT Machine Learning Club (UTMIST) *Toronto, Canada*

- Leading club website's migration to **Payload CMS**, replacing Google Forms and Git-push content updates with a single self-serve platform for **11 execs** and **100+ applicants** per year
- Interviewed all club VPs across engineering, research, academics, etc, to gather requirements for the website redesign, and combined input into a prioritized roadmap for the year.

Software Developer May 2026 – Aug 2026

Professor - University of Toronto *Toronto, Canada*

- Built a live queuing system using Next.j and SQL, reducing crowding during office hours for **500+ students**
- Integrated UofT SAML authentication for **15K+ students**, eliminating manual account creation
- Wrote GitHub Action to automatically publish updates via self hosted runner to save time during development
- Worked in a team of 4, meeting weekly with supervising professor to discuss progress and decide next steps

Projects

In Class Live QA Tool | *Next.js, Typescript, PostgreSQL, Redis, Docker*

- Architected a real-time UofT Q&A platform in **Next.js** securing a Fall rollout supporting **500+ students**
- Streamlined real time events using **WebSockets** to minimize delay in room events in **50+ student** rooms, achieving average latency of **40ms** during load test
- Implemented notifications and automatic question context with **React Hooks** for monitoring chats during lecture

SciContext – RAG Agent | *Python, PostgreSQL, AWS Bedrock*

- Built a RAG-powered query engine to help researchers parse scientific documents and research papers
- Implemented hybrid search using pgvector and BM25, resulting in an **85%** accuracy on a benchmark dataset
- Used section aware chunking to preserve additional context between headers and chunks
- Documents are chunked on markdown boundaries and indexed with meta data to better maintain semantic structure for better retrieval

Mysh – Linux Shell | *C, Systems Programming, POSIX*

- Built a C/POSIX Unix shell with pipelines, signal handling, and background-process management
- Implemented seven built-ins: echo, ls, cd, cat, wc, ps, and kill, mirroring core Unix workflows.
- Added TCP client/server commands supporting **50+ concurrent clients** across five milestones covering files, pipelines, signals, and networking.

Technical Skills

Languages: Typescript, C++, Java, Python
Frameworks: Next.js, React, Node.js, JUnit, Spring Boot, Agile
Developer Tools: Cursor, Codex, Git, Docker, PostgreSQL, Postman, Linux
Libraries: pytest, Prisma, PyTorch, JavaFX